FOR EDITH · PAEDIATRICS & CHILD HEALTH JUNIOR CLERKSHIP

Common Skin Diseases

Tutorial | Week 6 · Tetanus, skin, poisons & hormones | Paediatrics | Dr Mawanda

Skin disease is one of the commonest reasons Ugandan children attend clinic; most of it is infectious or infestation (scabies, tinea capitis, impetigo), atopic eczema or urticaria, all treatable with cheap topical or oral medicines, while a few skin signs are emergencies or clues to serious disease (Stevens–Johnson syndrome, HIV-related rashes, kwashiorkor dermatosis), and impetigo or infected scabies can lead to post-streptococcal glomerulonephritis.

Must know before the ward round

1. Describe before you diagnose: site and distribution, primary lesion (macule, papule, plaque, vesicle, bulla, pustule, wheal, nodule), secondary changes (scale, crust, excoriation, lichenification), colour, arrangement. Look at mouth, eyes, nails, hair and genitals.
2. Scabies: intense itch worse at night, burrows in finger webs, wrists, axillae, umbilicus, genitals; in infants palms, soles, face and scalp. Treat the WHOLE household at the same time and wash bedding.
3. Scabies treatment: permethrin 5% cream (from 2 months) or benzyl benzoate 25% (UCG 2023: neck-to-toe at bedtime, wash off in the morning, repeat; dilute 1:1 with water for ages 2–12, 1:3 under 2), repeat after 7 days; ivermectin 200 µg/kg orally, two doses 7 days apart, for children ≥15 kg.
4. Tinea capitis needs ORAL treatment (topicals cannot reach the hair follicle): griseofulvin with fatty food for 6–8 weeks (UCG 2023 10 mg/kg/day, doubled for severe; most international guides 15–25 mg/kg/day) or terbinafine by weight band for 4 weeks.
5. Tinea corporis: Whitfield's ointment or clotrimazole 1% twice daily, continued 2 weeks after clearing. Never use steroid creams on tinea (tinea incognito).
6. Impetigo: honey-coloured crusts (non-bullous) or flaccid blisters (bullous, Staph aureus). Few lesions: clean with chlorhexidine + gentian violet 0.5% or mupirocin. Widespread/fever: cloxacillin/flucloxacillin 12.5–25 mg/kg 6-hourly for 7 days (erythromycin 7.5–12.5 mg/kg 6-hourly if penicillin allergic).
7. Streptococcal skin infection (impetigo, infected scabies) can cause post-streptococcal glomerulonephritis 3–6 weeks later: check BP and urine for blood if a child later has oedema or dark urine.
8. Atopic eczema: emollients for everyone (liberal, several times a day; petroleum jelly), soap substitute, topical steroid by severity (hydrocortisone 1% for face/flexures; betamethasone 0.1% for body, 1–2 weeks), antihistamine for sleep, treat infection. Eczema herpeticum (punched-out erosions) is an emergency: aciclovir.
9. Fingertip unit (FTU) = about 0.5 g = enough steroid for an area of two adult palms.
10. Urticaria: wheals that each last <24 h; papular urticaria from insect bites is very common in Ugandan children. Non-sedating antihistamine (cetirizine 2.5 mg 1–6 years, 5 mg ≥6 years once daily, UCG). Lip/tongue swelling, wheeze or shock = anaphylaxis: IM adrenaline.
11. SJS/TEN: fever, painful dusky rash, blisters and erosions of TWO or more mucosae; SJS <10% body surface, TEN >30%. Commonest triggers in Uganda: cotrimoxazole, nevirapine, anticonvulsants (carbamazepine, phenytoin, phenobarbital, lamotrigine), anti-TB drugs. Stop the drug, treat like a burn, eye review, never rechallenge.
12. HIV clues in skin: papular pruritic eruption, extensive molluscum or warts, herpes zoster in a child, recurrent/severe fungal infection, severe seborrhoeic dermatitis, drug eruptions, Kaposi sarcoma. Test for HIV.

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In a nutshell

Skin complaints are among the **commonest reasons children attend outpatient clinics** in Uganda, and in schools and crowded homes some conditions (scabies, tinea capitis) affect a large share of children. Most are **not dangerous but are miserable**: itching ruins sleep and schooling, scratching causes secondary infection, and stigma is real.

A few skin conditions are **emergencies** or **windows to serious disease**:

- **Stevens–Johnson syndrome / toxic epidermal necrolysis (SJS/TEN)**: a drug reaction that can kill.
- **Eczema herpeticum**, **staphylococcal scalded skin**, **purpura/petechiae with fever** (meningococcaemia), **anaphylaxis** (urticaria with airway or circulation problems).
- **HIV-related skin disease**: often the first sign of HIV in a child.
- **Kwashiorkor dermatosis**: a sign of severe acute malnutrition.
- **Impetigo and infected scabies** can trigger **post-streptococcal glomerulonephritis** (and possibly rheumatic fever).

> **CLINICAL PEARL**
> The one thing never to forget: **describe the rash properly, then look at the whole child**: mucous membranes, temperature, nutrition, HIV status and the drug history. A rash with **mucosal erosions and a new drug** is SJS until proven otherwise.

Definitions & classification

Describing skin lesions (the language of dermatology)

Term	Meaning	Examples
Macule	Flat, colour change, <1 cm	Measles rash (early), vitiligo, café-au-lait spot
Patch	Flat, >1 cm	Pityriasis versicolor, large café-au-lait
Papule	Raised, solid, <0.5–1 cm	Scabies, molluscum, papular urticaria
Plaque	Raised, flat-topped, >1 cm	Psoriasis, chronic eczema, tinea corporis
Nodule	Solid, deep, >1 cm	Scabies nodules, erythema nodosum
Vesicle	Fluid-filled, <0.5 cm	Chickenpox, herpes, eczema (acute)
Bulla	Fluid-filled, >0.5 cm	Bullous impetigo, SJS/TEN, burns
Pustule	Pus-filled	Infected scabies, folliculitis
Wheal	Transient, oedematous, itchy plaque (<24 h)	Urticaria

Term	Meaning	Examples
Crust	Dried exudate	Impetigo (honey-coloured)
Scale	Flakes of keratin	Tinea, eczema, psoriasis
Erosion / ulcer	Partial / full-thickness loss of epidermis (and dermis)	SJS, eczema herpeticum / ecthyma, Buruli ulcer
Excoriation	Scratch mark	Any itchy condition
Lichenification	Thickened skin with exaggerated markings from rubbing	Chronic atopic eczema
Petechiae / purpura	Non-blanching red-purple spots, <2 mm / larger	Meningococcaemia, ITP, leukaemia, dengue

A useful order for presenting a rash: **S**ite and distribution → **C**olour → **A**rrangement (linear, annular, grouped, dermatomal) → **L**esion type (primary, then secondary) → **E**xtras (mucosae, nails, hair, nodes, systemic features). ("**SCALE**")

Classification of common childhood skin disease

Group	Conditions in this topic
Infestations	Scabies (also pediculosis, tungiasis/jiggers)
Fungal	Tinea capitis, corporis, pedis, unguium; pityriasis versicolor; candidiasis
Bacterial	Impetigo (non-bullous, bullous), ecthyma, folliculitis, boils, cellulitis, staphylococcal scalded skin syndrome
Viral	Molluscum contagiosum, warts, herpes simplex/zoster, chickenpox, measles
Inflammatory / allergic	Atopic eczema, seborrhoeic dermatitis, contact dermatitis, urticaria, papular urticaria
Drug reactions	Fixed drug eruption, morbilliform rash, **SJS/TEN**, DRESS
Nutritional	Kwashiorkor dermatosis ("flaky paint"), pellagra, zinc deficiency
HIV-related	Papular pruritic eruption, extensive molluscum/warts, zoster, seborrhoeic dermatitis, severe fungal infections, Kaposi sarcoma, drug eruptions

SJS/TEN definitions

Category	Body surface area with detachment
SJS	**<10%**
SJS/TEN overlap	**10–30%**
TEN	**>30%**

All have **mucosal involvement** (usually two or more sites: mouth, eyes, genitals) and are most often **drug-induced** (onset typically **4–28 days** after starting the drug).

Epidemiology & Uganda context

- Surveys of school children in Uganda and East Africa consistently find **tinea capitis, scabies, impetigo and pediculosis** among the commonest conditions, often in the same child. **Tinea capitis** is very common in primary-school children (spread by shared combs, hats and barbers' clippers).

- **Scabies** spreads in crowded homes, boarding schools, prisons, refugee settlements and institutions; WHO added scabies to the neglected tropical diseases list in 2017 and supports **mass drug administration with ivermectin** where prevalence is high. Infected scabies is a leading driver of **impetigo** and **post-streptococcal glomerulonephritis** in tropical countries.
- **Papular urticaria** (insect-bite hypersensitivity from mosquitoes, fleas, bedbugs) is very common in young children.
- **Atopic eczema** is rising in urban Uganda.
- **HIV**: skin disease affects most children living with HIV at some point; **papular pruritic eruption (PPE)** is a WHO clinical stage 2 condition and should prompt HIV testing.
- **SJS/TEN**: risk is much higher in people with HIV; in Uganda, cotrimoxazole prophylaxis, nevirapine (still used for infant prophylaxis), anti-TB drugs and anticonvulsants are the main triggers.

UGANDA CONTEXT

What is on the shelf at HC II–IV: **benzyl benzoate 25%** (permethrin is less often available and more expensive), **ivermectin** (in NTD programmes and some pharmacies), **Whitfield's ointment** (benzoic + salicylic acid), **clotrimazole/miconazole cream**, **griseofulvin tablets**, **gentian violet 0.5%**, **chlorhexidine**, **betamethasone 0.1% and hydrocortisone 1% creams**, **petroleum jelly (Vaseline)**, **chlorphenamine, cetirizine, promethazine**, **cloxacillin and erythromycin**, **calamine lotion**. Terbinafine tablets, mupirocin and permethrin are mostly in private pharmacies and hospitals. Dermatology services are concentrated at Mulago and a few regional referral hospitals.

Pathophysiology & principles

- **Skin barrier**: the stratum corneum keeps water in and irritants and microbes out. In **atopic eczema**, filaggrin and other barrier defects cause dry, leaky skin, allowing allergens and *Staph aureus* in, which drives type 2 (Th2) inflammation and itch; scratching damages the barrier further (**itch–scratch cycle**). This is why **emollients are the foundation** of treatment and why soap makes eczema worse.
- **Scabies**: the female *Sarcoptes scabiei* mite burrows into the stratum corneum and lays eggs. The **itch is a delayed hypersensitivity reaction** to mites, eggs and faeces, which is why it starts **3–6 weeks after first infestation** (1–3 days on re-infestation) and **persists for 2–4 weeks after successful treatment**. In immunosuppression (HIV, malnutrition, steroids) thousands of mites cause **crusted (Norwegian) scabies**, which is highly contagious.
- **Dermatophytes** (*Trichophyton, Microsporum*) digest keratin in skin, hair and nails. In **tinea capitis** they invade the **hair shaft and follicle**, out of reach of creams; hence oral treatment. An intense inflammatory reaction produces a boggy, pustular **kerion** (not an abscess: don't incise).
- **Impetigo**: *Staph aureus* (and *Strep pyogenes*) infect the superficial epidermis, often on top of scabies, eczema or insect bites. **Bullous impetigo** is caused by staphylococcal **exfoliative toxin**, which splits the epidermis; the same toxin circulating causes **staphylococcal scalded skin syndrome (SSSS)** in infants.
- **Post-streptococcal glomerulonephritis (PSGN)**: immune complexes from nephritogenic strains of group A streptococcus deposit in glomeruli, causing haematuria, oedema, hypertension and oliguria **3–6 weeks after a skin infection** (1–2 weeks after a throat infection).
- **Urticaria**: mast cells release histamine → leaky dermal vessels → wheals. Triggers in children: **viral infections** (commonest in acute urticaria), foods, drugs, insect bites, worms.
- **SJS/TEN**: a cytotoxic T-cell reaction to a drug causes widespread **keratinocyte death** (via granulysin and Fas ligand), so the epidermis peels off like a burn. HIV, genetic HLA types and slow drug metabolism increase risk.
- **Kwashiorkor dermatosis**: protein, zinc and essential fatty-acid deficiency and oxidative stress damage the epidermis: hyperpigmented patches that crack and peel ("flaky paint"), leaving raw, easily infected skin.

Clinical acumen

Focused history

Question	Why it matters
Onset, duration, where it started, how it spread	Acute (urticaria, drug reaction, viral) vs chronic (eczema, tinea)
Itch? Worse at night?	Severe nocturnal itch = scabies; itch is the hallmark of eczema and urticaria
Others at home or school with itch or rash	Scabies, tinea capitis, impetigo spread in households and schools
New medicines in the last 1–8 weeks (cotrimoxazole, nevirapine, anticonvulsants, anti-TB, antibiotics, herbal)	Drug eruption, SJS/TEN, DRESS
Fever, malaise, sore eyes or mouth	SJS, measles, chickenpox, sepsis, eczema herpeticum
Personal or family history of eczema, asthma, hay fever	Atopy
Soaps, creams, herbs, bleaching or steroid creams used	Contact dermatitis, tinea incognito, steroid damage
Insect exposure (bedbugs, fleas, mosquitoes, no net)	Papular urticaria
HIV status (child and mother), cotrimoxazole/ART	HIV-related skin disease; drug reactions
Feeding, weight loss, swelling of feet	Kwashiorkor dermatosis
Urine colour, facial swelling (recent impetigo/scabies)	PSGN
Immunisation (measles)	Measles rash and complications
Effect on sleep, school, family	Severity and impact

Examination

1. **The whole child**: temperature, nutrition (weight, MUAC, oedema), lymph nodes, pallor, hepatosplenomegaly, signs of HIV (oral thrush, parotid enlargement, lymphadenopathy).
2. **The whole skin in good light** (undress the child, with the mother present): distribution (flexures, extensor surfaces, sun-exposed, nappy area, palms and soles, scalp, finger webs).
3. **Primary lesion and secondary changes** (use the terms above).
4. **Mucous membranes**: mouth, lips, eyes (conjunctivitis), genitals: **crucial for SJS**.
5. **Hair, scalp, nails**: scaling, broken hairs, black dots, kerion (tinea capitis); nail dystrophy (fungal, psoriasis).
6. **Signs of infection**: crusts, pus, cellulitis, lymphangitis, fever.
7. **Nikolsky sign**: gentle sideways pressure on normal-looking skin shears off the epidermis = SSSS or TEN.
8. **Blood pressure and urine dipstick** in any child with recent impetigo/scabies and oedema or dark urine (PSGN).

Red flags

RED FLAG

- **Mucosal erosions + blisters/dusky skin + fever** after a new drug = **SJS/TEN**: stop the drug, admit.
- **Fever + non-blanching petechiae/purpura** = meningococcaemia/sepsis: ETAT, antibiotics now.
- **Eczema that suddenly becomes painful with clustered punched-out erosions and fever** = eczema herpeticum: IV/oral aciclovir.
- **Infant with fever, tender red skin, and peeling around the mouth and flexures** (positive Nikolsky, mucosa spared) = SSSS: IV cloxacillin/flucloxacillin.
- **Urticaria with lip/tongue swelling, stridor, wheeze or shock** = anaphylaxis: IM adrenaline.
- **Haematuria, oedema, hypertension** weeks after skin sores = PSGN.
- **Crusted, thick, scaly scabies** in an HIV-positive or malnourished child = crusted scabies: isolate, oral + topical treatment.
- **Flaky-paint skin with oedema** = kwashiorkor: SAM protocol.

Thinking like a clinician

- **Itch is the key branch point.** Itchy: scabies, eczema, urticaria/papular urticaria, tinea, PPE, lice. Not very itchy: impetigo, molluscum, viral exanthems, pityriasis versicolor, drug eruptions (variable).
- **"Is anyone else at home itching?"** If yes, think **scabies** and treat everyone.
- **Scaly patch on the scalp with hair loss in a school child = tinea capitis** until proven otherwise; topicals alone fail.
- **Ring-shaped rash treated with a steroid cream that got bigger and less typical** = **tinea incognito**.
- **Impetigo in a child with scabies**: treat both, otherwise the impetigo recurs.
- **A child on cotrimoxazole for HIV with fever and mouth sores** has SJS until proven otherwise; stop cotrimoxazole and never restart it.
- **Extensive molluscum, giant molluscum, or herpes zoster in a young child → test for HIV**.
- **Common student mistakes**: describing the rash vaguely ("a rash on the body"); not examining the mouth and eyes; treating only the child with scabies, not the household; topical antifungals for tinea capitis; potent steroids on the face or on fungal infections; antihistamines without emollients in eczema; missing a drug history.

EXAM FAVOURITE

Viva favourite: **"Why do children with scabies keep itching after treatment?"** The itch is a hypersensitivity reaction to dead mites and their products and can last **2–4 weeks** after successful treatment. Persistent itch with **new burrows** after 4 weeks, or itching household members, means treatment failure or re-infestation.

Clinical workup

Step 0: First 5 minutes

Most skin consultations are not emergencies, but always screen first:

1. **ABC**: stridor, wheeze, lip/tongue swelling (anaphylaxis); shock (sepsis, TEN with fluid loss).
2. **Fever + petechial/purpuric rash** → ETAT emergency: IV/IM ceftriaxone, fluids per protocol.
3. **Mucosal erosions + blistering/dusky skin** → SJS/TEN: stop suspect drugs, admit, assess fluid needs as for burns.

4. **Widespread tender red skin with peeling in an infant** → SSSS: admit, IV antibiotics.
5. **Signs of SAM** (oedema, wasting, flaky-paint skin) → SAM protocol (glucose, warmth, antibiotics).
6. **Eczema herpeticum** → aciclovir today.

> **RED FLAG**
> Call your senior for: SJS/TEN, SSSS, eczema herpeticum, a febrile child with purpura, anaphylaxis, or a rash in an immunocompromised child that is spreading quickly.

Step 1: Focused history

- [ ] **Onset, site of first lesion, spread, duration** → acute vs chronic pattern.
- [ ] **Itch and its timing** → scabies (nocturnal), eczema, urticaria.
- [ ] **Contacts with similar symptoms** → scabies, tinea capitis, impetigo.
- [ ] **All medicines, including herbal and creams, in the past 8 weeks** → drug reactions, steroid-modified tinea.
- [ ] **Fever, sore mouth or eyes, pain** → SJS, viral infection, eczema herpeticum.
- [ ] **Atopy** (personal, family) → eczema.
- [ ] **HIV status, ART/cotrimoxazole, immunisation, feeding/growth** → HIV skin disease, measles, kwashiorkor.
- [ ] **Previous treatment and response** → resistance, wrong diagnosis, poor adherence, re-infestation.

Step 2: Focused examination

General: temperature, weight, MUAC, oedema, lymph nodes, signs of HIV.

Skin: distribution, primary lesion, secondary change, colour, arrangement; **scalp, nails, finger webs, genitals, palms and soles**; **mouth and eyes**; Nikolsky sign if blistering.

Document these positives	And these negatives
Distribution and lesion type ("excoriated papules and burrows in the finger webs, wrists and periumbilical area")	"Mucous membranes normal"
Secondary infection (crusts, pustules), lymph nodes	"No fever; no purpura"
Scalp: scale, broken hairs, kerion	"Nikolsky sign negative"
Nutrition: weight, MUAC, oedema	"No oedema; BP normal; urine dipstick negative for blood"

Step 3: Bedside tests

- **Skin scraping with 10–20% KOH** (scale from the active edge, or plucked hairs) → fungal hyphae/spores (tinea).
- **Scabies**: dermatoscopy or a burrow scraping with mineral oil under the microscope (mites, eggs, faeces); usually a clinical diagnosis.
- **Wood's lamp**: green fluorescence of *Microsporum* tinea capitis (most *Trichophyton* species do not fluoresce).
- **Urine dipstick and BP**: PSGN after skin infection.
- **Glucose**: recurrent boils/candida (diabetes).
- **Rapid HIV test**: HIV clues.

Step 4: Laboratory & imaging

Test	When	Expected / abnormal finding	How it changes management	Uganda availability
KOH microscopy	Tinea, pityriasis versicolor, candida	Hyphae, spores; "spaghetti and meatballs" in versicolor	Confirms fungal; choose oral vs topical	HC IV and hospitals with microscopy
Fungal culture (Sabouraud's)	Treatment failure, outbreaks	Species	Choice of griseofulvin vs terbinafine	National/teaching labs
Pus swab Gram stain/culture	Severe impetigo, abscess, SSSS	*S. aureus* (MRSA?), *S. pyogenes*	Antibiotic choice	Hospitals
Skin scraping for mites	Doubtful scabies, crusted scabies	Mites, eggs, scybala	Confirms	Hospitals
HIV test	PPE, extensive molluscum/warts, zoster, SJS, severe or recurrent infection	Positive	ART, cotrimoxazole, staging	Everywhere
FBC, eosinophils, stool microscopy	Chronic urticaria, prurigo	Eosinophilia, helminths	Deworm	Hospitals
Urinalysis, creatinine, ASOT, C3	Oedema/haematuria after skin infection	Haematuria, proteinuria, raised ASOT, low C3	PSGN management	RRH/national referral
FBC, electrolytes, albumin, culture	SJS/TEN	Fluid/electrolyte loss, sepsis	Fluids, antibiotics if infected	Hospitals
Skin biopsy	Uncertain diagnosis, suspected KS, TEN vs SSSS	Histology	Diagnosis	Mulago and a few RRHs

Step 5: Putting it together

Model one-line summary: "A 7-year-old boy, 21 kg, with 3 weeks of intensely itchy excoriated papules and burrows in the finger webs, wrists, axillae and genitals, secondary impetiginisation of the hands, and two itchy siblings: **scabies with secondary impetigo**; BP and urine normal."

> NOTE
>
> **Worked example: is it tinea or eczema?** A 9-year-old has a 3 cm round, scaly, itchy plaque on the forearm with a **raised active edge and central clearing**; KOH shows hyphae. **Interpretation**: **tinea corporis** (annular with an active scaly edge; eczema is ill-defined without central clearing and is usually symmetrical in flexures). **Plan**: Whitfield's ointment or clotrimazole 1% twice daily until 2 weeks after clearing; check the scalp and household; no steroid cream.

Differential diagnosis

The itchy child

Condition	Distinguishing features	Key test
Scabies	Night itch, burrows, finger webs, wrists, genitals, family affected; infants: palms, soles, scalp	Clinical; mite scraping

Condition	Distinguishing features	Key test
Atopic eczema	Dry skin; infants: face and extensor; children: flexures (elbows, knees, neck); chronic and relapsing; atopy	Clinical
Papular urticaria	Grouped itchy papules on exposed skin (legs, arms), central punctum, new crops after bites	Clinical; no burrows, family members less affected
Papular pruritic eruption (HIV)	Symmetrical, very itchy papules on limbs and trunk, hyperpigmented scars	HIV test
Tinea corporis	Annular scaly plaque with active edge	KOH
Pediculosis	Scalp itch, nits on hair shafts	Look for lice/nits
Onchocerciasis (endemic foci)	Chronic itchy papular dermatitis, depigmentation ("leopard skin"), nodules, eye disease	Skin snip

Blistering or peeling child

Condition	Distinguishing features	Key test
SJS/TEN	Drug history, dusky target-like lesions, blisters, **mucosal erosions (≥2 sites)**, Nikolsky positive in TEN	Clinical ± biopsy
SSSS	Infants, fever, tender erythema, superficial peeling around mouth and flexures, **mucosa spared**	Clinical; swab from focus
Bullous impetigo	Localised flaccid bullae with collarette of scale, well child	Swab
Chickenpox	Crops of vesicles in different stages, central distribution, mild mucosal	Clinical
Eczema herpeticum	Punched-out, clustered erosions on eczema, fever	Clinical (Tzanck/PCR)
Burns (including scalds, abuse)	History, pattern	History; safeguarding

Scaly scalp

Condition	Distinguishing features	Key test
Tinea capitis	Patchy hair loss, broken hairs, black dots, scale, lymph nodes, kerion	KOH of hairs, Wood's lamp
Seborrhoeic dermatitis	Infants (cradle cap) or adolescents; greasy yellow scale; no hair loss	Clinical
Atopic eczema	Other eczema sites, itch	Clinical
Psoriasis	Thick silvery plaques at hairline, nails pitted	Clinical
Traction alopecia	Tight braiding, hair loss at margins	History

Investigations

Test	What you are looking for	Why
KOH microscopy	Fungal elements	Confirms tinea before weeks of griseofulvin
Pus swab	Organism and sensitivity	Severe or recurrent infection
HIV test	HIV infection	Many skin signs are HIV-associated; affects treatment
Urine dipstick, BP	PSGN	After streptococcal skin infection
FBC, eosinophils, stool	Helminths, chronic urticaria	Deworming

Test	What you are looking for	Why
Skin biopsy	Uncertain or serious diagnoses	KS, TEN vs SSSS

Management

A. General principles of skin treatment

1. **Treat the cause** (mite, fungus, bacteria, drug, allergen) and **the itch** (antihistamine, emollient).
2. **Choose the right vehicle**: **ointments** (greasy) for dry, scaly skin; **creams** for moist areas and the face; **lotions** for hairy areas and large areas; **pastes** for protection.
3. **Quantity**: one **fingertip unit (FTU)** (about 0.5 g) covers two adult palms. A child's whole body needs about 10–20 FTUs per application depending on age.
4. **Treat contacts** (scabies, tinea capitis, impetigo in households and schools).
5. **Prevent**: hygiene, not sharing combs, hats, towels and bedding; sleeping under nets; nutrition.

B. Scabies

General measures: treat **all household members and close contacts at the same time**, whether or not they itch; wash bedding, clothes and towels in hot water and dry in the sun (or seal in a bag for 72 hours); cut nails short.

Drug	Dose / application	Age limits	Frequency / duration	Notes
Permethrin 5% cream (first-line internationally)	Apply to the **whole body from the neck down** (include scalp, face and ears in infants and young children, avoiding eyes and mouth), especially finger webs, under nails, genitals; wash off after **8–12 hours**	From **2 months**	**Repeat after 7 days**	Reapply to hands if washed during treatment
Benzyl benzoate 25% lotion (UCG 2023 first line, widely available)	After a bath, apply to the whole body, bedtime, wash off in the morning (UCG: "repeat 2 times")	**Dilute for children**: 1 part lotion + 1 part water for **2–12 years**; 1 part lotion + 3 parts water **under 2 years** (UCG dilutions); avoid in young infants if permethrin or sulphur is available	Repeat; many protocols repeat after 7 days to kill newly hatched mites	Stings and irritates; protect eyes; not in pregnancy per UCG
Sulphur 5–10% ointment	Whole body nightly	**Infants <2 months**, pregnancy	**3 consecutive nights**, wash off 24 h after last application	Messy, smelly, but safe
Ivermectin	**200 µg/kg** orally with food	Children **≥15 kg** (UCG also says avoid under 12 years where possible); not in pregnancy	**Two doses 7 days apart**	For crusted scabies, outbreaks, failure of topicals; combine with topical in crusted scabies

Drug	Dose / application	Age limits	Frequency / duration	Notes
Chlorphenamine (itch)	**1–2 mg** per dose (UCG); 1–2 years 1 mg 12-hourly; 2–5 years 1 mg 6-hourly; 5–12 years 2 mg 6-hourly	From 1 year	For 3–7 days, mainly at night	Sedating; useful for sleep
Antibiotic for secondary infection	See impetigo	—	7 days	Treat infection and scabies together

CLINICAL PEARL

Tell families **the itch can last 2–4 weeks after treatment**; give an antihistamine and emollient, and review at 2–4 weeks. **New burrows** at that point mean failure or re-infestation (usually because contacts were not treated).

Crusted scabies (HIV, SAM, steroids, disability): isolate; **oral ivermectin 200 µg/kg on days 1, 2 and 8** (more doses in severe cases) **plus** topical permethrin/benzyl benzoate repeatedly; keratolytic (salicylic acid in petroleum jelly) to remove crusts; treat contacts; HIV test.

C. Fungal infections

Tinea capitis

Drug	Dose	Route	Frequency / duration	Notes
Griseofulvin (microsize)	**UCG 2023: 10 mg/kg/day** (double in severe infection); international guides **15–25 mg/kg/day** (max 1 g/day)	Oral, **with fatty food** (milk, groundnuts, avocado)	Once daily (or divided) for **6–8 weeks** (UCG 6 weeks), until clinically and mycologically clear	Best for *Microsporum*; headache, GI upset, photosensitivity; avoid in pregnancy/SLE/porphyria
Terbinafine	**<20 kg: 62.5 mg; 20–40 kg: 125 mg; >40 kg: 250 mg**	Oral	Once daily for **4 weeks** (*Trichophyton*)	Better for *Trichophyton*; check LFTs if prolonged
Adjuncts	Ketoconazole 2% or selenium sulphide shampoo	Topical	Twice weekly for 2–4 weeks	Reduces spore shedding; treat household contacts with shampoo
Kerion	Oral antifungal as above ± short course oral prednisolone (1 mg/kg for 1–2 weeks) for severe inflammation	—	—	**Do not incise**; antibiotics only if secondary infection

DRUG DOSING

Do NOT treat tinea capitis with creams alone: they cannot reach the infected hair follicle (UCG 2023). Children can stay in school once treatment has started, but should not share combs, hats or clippers.

Tinea corporis, pedis, cruris

- **Whitfield's ointment** (benzoic acid 6% + salicylic acid 3%) **12-hourly until 2 weeks after the lesions clear** (UCG), or **clotrimazole 1% / miconazole 2% cream** twice daily for 2–4 weeks.
- **Oral griseofulvin** (as above) for 3–6 weeks if extensive or not responding to topicals.
- **Never use a steroid cream** (tinea incognito); combined steroid–antifungal–antibiotic creams sold over the counter cause this.

Pityriasis versicolor

- Hypo- or hyperpigmented, finely scaly macules on the trunk (adolescents); *Malassezia*.
- **Clotrimazole/miconazole cream 12-hourly for 2–3 weeks**, or **selenium sulphide 2.5% / ketoconazole 2% shampoo** applied to the skin for 10 minutes daily for 1–2 weeks; oral fluconazole for extensive disease in adolescents (UCG: 300 mg weekly for 2 weeks); **not griseofulvin** (ineffective).
- Pigment takes months to return; that is not failure.

Candidiasis (nappy area, oral)

- Nappy candidiasis: bright red with satellite papules; nystatin or clotrimazole cream 2–3 times daily, keep the area dry, zinc barrier cream.
- Oral thrush: nystatin suspension 100,000 units 6-hourly for 7 days (infants 1 mL); persistent or severe thrush beyond infancy → HIV test.

D. Impetigo and other bacterial infections

Severity	Treatment	Dose	Duration
Localised (<5 lesions), well child (UCG)	Soak and gently remove crusts with soap and water; clean with **chlorhexidine 0.05%**; **gentian violet 0.5%** 12-hourly **or mupirocin 2%** 8-hourly (or fusidic acid 2%)	Topical	Gentian violet 3 days; mupirocin 5–7 days
Widespread (>5 lesions), bullous, fever or nodes	**Cloxacillin/flucloxacillin**	**12.5–25 mg/kg 6-hourly** (max 500 mg/dose), before food	**7 days**
Penicillin allergy	**Erythromycin**	**7.5–12.5 mg/kg 6-hourly** (UCG 7.5 mg/kg)	7 days
Cellulitis needing admission	IV cloxacillin 25–50 mg/kg 6-hourly ± benzylpenicillin; or ceftriaxone 50 mg/kg daily	IV	Switch to oral when improving; total 7–10 days
SSSS	IV cloxacillin/flucloxacillin 50 mg/kg 6-hourly, fluids, skin care as a burn, analgesia	IV	7–10 days

Also: treat underlying scabies, eczema or lice; keep nails short; separate towels; children can return to school once lesions have crusted over or 48 hours after starting antibiotics; **check urine and BP if the child later develops oedema or dark urine (PSGN)**.

E. Atopic eczema (NICE CG57 stepped care; UCG 2023)

Severity	What you see	Treatment
Mild	Dry skin, occasional itch, small areas of redness	**Emollients** + **mild steroid** (hydrocortisone 1%) for flares
Moderate	Dry skin, frequent itch, redness ± excoriation and localised thickening	Emollients + **moderate steroid** (e.g. clobetasone butyrate 0.05%) for flares; **hydrocortisone 1% for face/neck/flexures**; sedating antihistamine at night for 1–2 weeks if sleep disturbed
Severe	Widespread dryness, incessant itch, redness, excoriation, extensive thickening, bleeding, oozing, cracking	Emollients + **potent steroid** (betamethasone valerate 0.1%) for the body for up to 1–2 weeks (UCG: 12-hourly for 2 weeks, not face/genitals); wet wraps; refer if not controlled

Practical regimen:

1. **Emollient**: petroleum jelly, emulsifying ointment or a cream, **liberally (250–500 g per week in a child), 2–4 times a day and after bathing**; continue when the skin is clear.
2. **Soap substitute**: wash with emollient, not soap; lukewarm water; pat dry; bathe at most once or twice daily (UCG).
3. **Topical steroid** once or twice daily to **active eczema only**, in FTUs, for **3–14 days** per flare; step down as it improves; potent steroids only for short periods and **not on the face or genitals** (hydrocortisone 1% there).
4. **Antihistamine**: sedating (chlorphenamine, promethazine 1 mg/kg/day in 1–2 doses, UCG; avoid promethazine under 2 years) for nights when itch disrupts sleep; non-sedating antihistamines add little for eczema itch.
5. **Infected eczema** (weeping, crusting, pustules, sudden worsening): oral **flucloxacillin/cloxacillin** 12.5–25 mg/kg 6-hourly for 7 days.
6. **Eczema herpeticum** (painful clustered punched-out erosions, fever): **aciclovir** oral 20 mg/kg (max 400 mg) 5 times a day for 5–7 days, or **IV 10–20 mg/kg 8-hourly** if unwell or periocular involvement (eye review).
7. **Triggers**: soaps, wool and synthetic clothes, heat and sweating, dust; food allergy is an uncommon cause except in infants with severe early eczema.
8. **Education**: chronic relapsing condition; most children improve with age; steroid fears ("steroid phobia") are common, so explain safe use with FTUs.

F. Molluscum contagiosum

- Pearly, dome-shaped, **umbilicated papules** 1–5 mm, often in clusters; spread by contact and autoinoculation; poxvirus.
- **Usually self-limiting over 6–18 months**; reassure; avoid squeezing and sharing towels.
- Treat if numerous, distressing or infected: curettage, cryotherapy, or gentle expression (specialist/clinic); topical potassium hydroxide 5–10% or podophyllotoxin in older children (not widely available).
- **Extensive, large (giant) or facial molluscum, especially in an older child = test for HIV**; lesions regress with **ART**.

G. Urticaria and papular urticaria

Drug	Dose	Route	Frequency	Notes
Cetirizine (non-sedating; first choice)	**1–6 years 2.5 mg; ≥6 years 5 mg** (UCG; 10 mg for adolescents)	Oral	Once daily (or 2.5–5 mg twice daily per BNFc)	Continue for days to weeks until wheals stop
Chlorphenamine (sedating)	1–2 years 1 mg 12-hourly; 2–5 years 1 mg 6-hourly; 5–12 years 2 mg 6-hourly (UCG)	Oral	As above	Night-time use
Promethazine	**1 mg/kg/day** in 1–2 doses (UCG)	Oral	—	**Not under 2 years** (respiratory depression)
Prednisolone (severe/unresponsive)	**1 mg/kg**	Oral	Once daily for **3–5 days** (UCG)	Short course only
Calamine lotion	Topical	—	2–3 times daily	Symptomatic relief
Adrenaline 1:1000	**0.01 mL/kg IM** (max 0.5 mL)	IM thigh	Every 5 min	If angioedema with airway, breathing or circulation compromise (anaphylaxis)

- **Acute urticaria** (<6 weeks): usually viral infection, food, drug, insect; avoid the trigger; antihistamines.

- **Papular urticaria**: crops of itchy papules on exposed skin after insect bites; **bed nets, treat bedbugs/fleas** (wash bedding, sun mattresses), antihistamine, mild topical steroid for a few days, treat infected lesions; usually outgrown by school age.
- **Chronic urticaria** (>6 weeks): look for worms (deworm with albendazole), thyroid disease; refer.

H. Stevens–Johnson syndrome / TEN

1. **Stop the culprit drug (and all non-essential drugs)** immediately; record it as an allergy on the file, card and in the child's health book; **never rechallenge**.
2. **Admit** (ideally to a high-dependency or burns area; refer TEN to Mulago/RRH).
3. **Fluids**: like a burn (losses through denuded skin): maintenance plus replacement guided by urine output (target ≥1 mL/kg/h), electrolytes and albumin.
4. **Skin care**: leave blisters in place where possible, **non-adherent dressings** (paraffin gauze, petroleum jelly), barrier nursing, warm room, handle gently; avoid adhesive tapes.
5. **Eyes**: ophthalmology review **within 24 hours**; lubricant drops, saline cleaning, antibiotic eye ointment, separate lids to prevent adhesions (blindness is a major long-term complication).
6. **Mouth and genitals**: saline/chlorhexidine mouthwashes, petroleum jelly to lips, analgesia before feeds; genital care (petroleum jelly, separate labia) to prevent adhesions.
7. **Nutrition**: NG feeds if unable to eat.
8. **Analgesia**: paracetamol, morphine as needed.
9. **Infection**: monitor; antibiotics only for proven or strongly suspected infection (cultures); no prophylactic antibiotics.
10. **Steroids**: **no strong evidence** (UCG 2023), and they increase infection risk; specialist decision only.
11. **HIV**: test; if on cotrimoxazole prophylaxis, **switch to dapsone 2 mg/kg once daily (max 100 mg)** after recovery; if nevirapine was the cause, switch ART per HIV guidelines (DTG-based).

I. HIV-related skin disease

Condition	Features	Management
Papular pruritic eruption (WHO stage 2)	Symmetrical intensely itchy papules on limbs and trunk, post-inflammatory hyperpigmentation	**ART** (main treatment), antihistamine, emollient, mild–moderate topical steroid, insect avoidance
Seborrhoeic dermatitis	Greasy scale on scalp, face, chest; severe and extensive in HIV	Ketoconazole 2% shampoo/cream, mild steroid, ART
Extensive molluscum, warts (stage 2)	Many or giant lesions	ART; local treatment
Herpes zoster (stage 2 in children)	Painful dermatomal vesicles; recurrent or multi-dermatomal	**Aciclovir 20 mg/kg (max 800 mg) 5 times daily for 7 days** (IV 10 mg/kg 8-hourly if disseminated or ophthalmic), analgesia, ART
Chronic herpes simplex >1 month (stage 4)	Persistent painful ulcers	Aciclovir, ART
Fungal infections (extensive tinea, nail infections, candida)	Severe, recurrent	Oral antifungals, ART
Kaposi sarcoma (stage 4)	Purple macules, plaques or nodules on skin, palate; lymphadenopathy	Refer to Uganda Cancer Institute; ART + chemotherapy
Drug eruptions	Cotrimoxazole, nevirapine, anti-TB drugs, efavirenz	Assess severity; SJS → stop; mild → may continue with antihistamine under review

Condition	Features	Management
Crusted scabies	Thick crusts, little itch, highly contagious	Ivermectin + topical, isolation

> **UGANDA CONTEXT**
> Under Uganda's consolidated HIV guidelines, every child with an HIV-associated skin condition (PPE, extensive molluscum or warts, zoster) should be **tested for HIV** and, if positive, started on **ART promptly** (DTG-based regimens) with **cotrimoxazole prophylaxis** unless there is a sulfa reaction. Skin often improves dramatically on ART.

J. Kwashiorkor dermatosis (WHO SAM protocol)

- **Appearance**: hyperpigmented, cracked, peeling patches ("**flaky-paint**" or "crazy-paving" dermatosis), raw depigmented areas, especially in the nappy area, legs and pressure areas; hair thin, sparse, light-coloured (flag sign); oedema.
- **Management** (inside the SAM protocol: glucose, warmth, antibiotics, F-75, micronutrients including **zinc**):
  - Leave exposed areas to dry; avoid nappies if the perineum is raw.
  - **Barrier cream** (zinc and castor oil ointment, or petroleum jelly) on raw areas; paraffin gauze dressings to raw areas.
  - **Bathe or soak** affected areas in **0.01% potassium permanganate** for 10 minutes daily (WHO) to dry and disinfect.
  - **Gentian violet** or **nystatin cream** for skin sores and candida.
  - Systemic antibiotics are already part of complicated SAM treatment.
- Skin heals as nutrition recovers; peeling worsening in the first days can also be a sign of **refeeding/zinc deficiency**: ensure the mineral mix is in the feeds.

K. Setting-specific plan

Level	What you can do
HC II/III	Scabies (benzyl benzoate, family treatment), tinea corporis (Whitfield's), tinea capitis (griseofulvin), impetigo (gentian violet, cloxacillin), eczema (emollient, hydrocortisone/betamethasone), urticaria (antihistamine), HIV testing; **refer** SJS/TEN, SSSS, eczema herpeticum, extensive or unclear disease
HC IV	As above plus admission for cellulitis, IV antibiotics, KOH microscopy
RRH / national referral	SJS/TEN care, dermatology review, biopsy, ophthalmology, burns-type care, KS referral

Complications

Condition	Complications
Scabies	Secondary impetigo, cellulitis, **PSGN**, possibly rheumatic fever, sepsis; sleep loss, school absence; crusted scabies in the immunosuppressed
Tinea capitis	Kerion, scarring alopecia (permanent hair loss), id reaction, spread in schools
Impetigo	Cellulitis, ecthyma, lymphangitis, **PSGN**, sepsis, SSSS (staphylococcal)
Atopic eczema	Secondary bacterial infection, **eczema herpeticum**, sleep disturbance, poor school performance, growth effects from severe disease, steroid side-effects (skin atrophy, striae) with misuse

Condition	Complications
Urticaria	Angioedema, anaphylaxis
SJS/TEN	Sepsis, dehydration, hypothermia, **blindness** and corneal scarring, oesophageal and genital strictures, nail loss, pigment change, death (TEN mortality is high)
HIV-related skin	Disfigurement, stigma, severe infections; KS can be fatal
Kwashiorkor skin	Infection, sepsis, hypothermia, fluid and protein loss

Ward-round & exam pearls

Model case presentation

> "Master K.M., a **6-year-old boy** from Kisenyi, **weight 18 kg** (weight-for-age Z –1.2, height-for-age Z –1.0), brought with **3 weeks of intense itching, worse at night**, which kept him and his two younger sisters awake; his mother is also itching. He has had **painful crusted sores on both hands for 5 days**. No fever, no recent new medicines, no oedema or dark urine. **Immunisation complete for age**, **eats family food**, **HIV-negative** (tested at 18 months; mother HIV-negative). Examination: well, afebrile; **excoriated papules and burrows in the finger webs, wrists, anterior axillary folds, periumbilical area and penis**, with **honey-coloured crusts over the dorsum of both hands** and tender epitrochlear nodes; scalp, mouth and eyes normal; **BP 98/60, urine dipstick negative for blood**. **Assessment: scabies with secondary impetigo**, household outbreak. **Plan**: benzyl benzoate 25% diluted 1:1 with water for him and his 4-year-old sister, 1:3 for the 18-month-old, full strength for the mother (or permethrin 5% for all) applied the same night and repeated; wash bedding; **cloxacillin 250 mg 6-hourly for 7 days** (about 14 mg/kg); chlorphenamine 2 mg at night; review in 2 weeks; return if facial swelling or dark urine."

Viva questions with answers

1. **Where do you look for scabies burrows?** Finger webs, sides of fingers, wrists, elbows, anterior axillary folds, nipples, umbilicus, buttocks, genitals (penis/scrotum); in infants also palms, soles, face and scalp.
2. **Why treat the whole household?** Contacts are infested before they itch (incubation 3–6 weeks), so untreated contacts re-infest the child.
3. **Why oral treatment for tinea capitis?** The fungus lives in the hair shaft and follicle, which topicals cannot penetrate.
4. **Name three drugs that commonly cause SJS in Ugandan children.** Cotrimoxazole (sulfonamides), nevirapine, carbamazepine/phenytoin/phenobarbital/lamotrigine, anti-TB drugs.
5. **What is a fingertip unit?** The amount of cream squeezed from a tube with a 5 mm nozzle along an adult index finger from tip to first crease, about 0.5 g, enough for two adult palms.
6. **What skin findings suggest HIV in a child?** Papular pruritic eruption, extensive molluscum or warts, herpes zoster, severe seborrhoeic dermatitis, recurrent severe fungal infections, Kaposi sarcoma, severe drug reactions.
7. **What renal complication follows impetigo?** Post-streptococcal glomerulonephritis 3–6 weeks later: haematuria, oedema, hypertension, oliguria.

OSCE tips

- In a "describe this rash" station, use the **SCALE** order and proper terms; end with "I would like to examine the mucous membranes, nails, hair, lymph nodes, and check the temperature, BP and urine".

- In a counselling station on scabies: everyone at once, neck to toe (whole body in infants), leave on overnight, repeat in a week, wash bedding, itch can last 2–4 weeks.
- Demonstrate an **FTU** and explain emollient use ("soak and smear").

How to present this tutorial

> NOTE
> **Edith presents this tutorial** with **Ampaire Perez**, moderated by **Dr Mawanda** (Week 6, Tuesday, 08:00–09:30). Skin is a visual subject: the success of the tutorial depends on **good pictures** (from the ward, with consent, or from open-access atlases such as DermNet) and on teaching colleagues to **describe** lesions, not just name them.

Plan the 90 minutes

Time	Segment	Who	What to do
0–5 min	**Picture quiz opener**	Edith	Show 3 photos (scabies hand, tinea capitis, honey-crusted impetigo) and ask the group to describe, not diagnose
5–15 min	**How to describe a rash**	Edith	Lesion vocabulary table; SCALE order; Nikolsky sign; "always look in the mouth"
15–30 min	**Infestations and infections**	Ampaire	Scabies (household treatment, dilutions, ivermectin), tinea capitis/corporis/versicolor, impetigo (topical vs oral, PSGN link), molluscum
30–45 min	**Inflammatory and allergic**	Edith	Atopic eczema (emollients, steroid ladder, FTU, eczema herpeticum), urticaria and papular urticaria (antihistamine doses, when it is anaphylaxis)
45–60 min	**The serious ones**	Ampaire	SJS/TEN (drugs, % BSA, burn-like care, eyes, dapsone switch), HIV-related skin disease (PPE, zoster, molluscum, KS), kwashiorkor dermatosis
60–75 min	**Cases and quiz**	Both	Two short cases from this page; 5 rapid MCQs
75–90 min	**Tutor discussion**	Dr Mawanda	Questions; tutor's emphasis

Suggested slides (14–16)

1. Title, names, date, tutor.
2. Picture quiz (three photos, "describe it").
3. Why skin matters in Ugandan children (school surveys, PSGN, HIV, SJS).
4. Describing lesions: vocabulary table.
5. The itchy child: differential table.
6. Scabies: picture, burrow sites, treatment table, household rule.
7. Tinea capitis: picture, why oral, griseofulvin/terbinafine doses.
8. Tinea corporis and versicolor; no steroids on fungus.
9. Impetigo: picture, topical vs oral, PSGN.
10. Molluscum and the HIV clue.
11. Atopic eczema: stepped care, FTU, eczema herpeticum warning.
12. Urticaria and papular urticaria; anaphylaxis box.
13. SJS/TEN: drugs, % BSA, management checklist.
14. HIV-related skin disease table.

15. Kwashiorkor skin and SAM care.
16. Five take-home messages.

> CLINICAL PEARL
> **Bring props**: a tube of cream to demonstrate a **fingertip unit**, a bottle of benzyl benzoate and a measuring cup to show the **1:1 and 1:3 dilutions**, a tub of petroleum jelly (the cheapest effective emollient), and a printed one-page handout with the drug tables. Ask permission before photographing any patient, and remove identifying features.

Questions Dr Mawanda is likely to ask (prepare answers)

- **"How do you dilute benzyl benzoate for a 3-year-old and a 1-year-old?"** 1:1 with water (2–12 years); 1:3 under 2 years (UCG); or use permethrin 5% (from 2 months) or sulphur ointment in young infants.
- **"Why did this child's tinea get worse with the cream from the pharmacy?"** It was a steroid-containing combination cream: steroid suppresses inflammation and lets the fungus spread (tinea incognito).
- **"When would you give oral antibiotics for impetigo?"** More than about five lesions, bullous impetigo, fever, lymphadenitis, or failure of topical treatment: cloxacillin 12.5–25 mg/kg 6-hourly for 7 days.
- **"A child on cotrimoxazole develops mouth ulcers and a blistering rash. What do you do?"** Stop cotrimoxazole, admit, manage as SJS (fluids, skin, eye, mouth care), HIV review, switch prophylaxis to dapsone after recovery, never rechallenge.
- **"How do you know a child's eczema is infected, and what do you give?"** Weeping, crusting, pustules, sudden worsening, fever; flucloxacillin/cloxacillin; if punched-out painful erosions, aciclovir for eczema herpeticum.
- **"Which skin conditions are WHO stage 2 in children?"** Papular pruritic eruption, extensive molluscum, extensive wart virus infection, fungal nail infections, herpes zoster, recurrent oral ulcers, lineal gingival erythema, angular cheilitis.

Delivery tips

- Show a picture, **pause**, and let the audience describe it before you speak.
- Keep doses on screen with **weights** ("for an 18 kg child…").
- End with **five take-home messages**: (1) describe first, then diagnose, and always look in the mouth; (2) scabies: treat the whole household at once and repeat in a week; (3) tinea capitis needs oral treatment; never steroids on fungus; (4) emollients are the backbone of eczema care; (5) mucosal erosions + new drug = SJS: stop the drug.

Mnemonics & memory aids

- **Describe a rash: "SCALE"**: **S**ite/distribution, **C**olour, **A**rrangement, **L**esion (primary then secondary), **E**xtras (mucosa, nails, hair, nodes, systemic).
- **Scabies sites: "Webs, Wrists, Waist, Willy, Women's nipples"** (plus palms, soles and scalp in babies).
- **Scabies rule "ALL-AT-ONCE, NECK-TO-TOE, REPEAT-IN-SEVEN"**.
- **Tinea capitis "Hair needs a pill"**.
- **Impetigo "Honey crusts → think Strep → think kidneys"**.
- **SJS triggers in Uganda "CANNA"**: **C**otrimoxazole, **A**nticonvulsants, **N**evirapine, **N**SAIDs, **A**nti-TB drugs (and allopurinol).
- **SJS sizes "10–30"**: <10% SJS, 10–30% overlap, >30% TEN.
- **Eczema "Soak and smear"**: bathe in lukewarm water, then emollient on damp skin; steroid only on red, itchy areas.

High-yield summary

Numbers & doses to know

Item	Value
Permethrin 5%	Neck to toe (whole body in infants), 8–12 h, **repeat after 7 days**, from **2 months**
Benzyl benzoate 25%	Overnight, repeat; **1:1 dilution 2–12 years; 1:3 under 2 years** (UCG)
Ivermectin (scabies)	**200 µg/kg**, 2 doses **7 days apart**, child **≥15 kg**
Scabies itch after treatment	Up to **2–4 weeks**
Griseofulvin (tinea capitis)	UCG **10 mg/kg/day** (double if severe); international 15–25 mg/kg/day; **6–8 weeks**, with fatty food
Terbinafine (tinea capitis)	<20 kg **62.5 mg**; 20–40 kg **125 mg**; >40 kg **250 mg** daily × **4 weeks**
Whitfield's / clotrimazole (tinea corporis)	Twice daily until **2 weeks after clearing**
Impetigo topical	Gentian violet 0.5% 12-hourly × 3 days; mupirocin 2% 8-hourly × 5–7 days
Cloxacillin/flucloxacillin	**12.5–25 mg/kg 6-hourly × 7 days**
Erythromycin	**7.5–12.5 mg/kg 6-hourly × 7 days**
PSGN after skin infection	**3–6 weeks**
Fingertip unit	**0.5 g** = two adult palms
Emollient amount	**250–500 g/week** for a child
Cetirizine	**2.5 mg (1–6 y); 5 mg (≥6 y)** daily (UCG)
Chlorphenamine	**1–2 mg** per dose (UCG bands)
Prednisolone (severe urticaria)	**1 mg/kg/day × 3–5 days**
SJS / overlap / TEN	**<10% / 10–30% / >30%** BSA
Dapsone (PCP prophylaxis after cotrimoxazole SJS)	**2 mg/kg daily** (max 100 mg)
Aciclovir (zoster, eczema herpeticum)	**20 mg/kg** (max 400–800 mg) oral 5 times daily; IV 10–20 mg/kg 8-hourly if severe

Classic exam traps

- **"Treat the child with scabies only."** Wrong: treat **the whole household at the same time**.
- **"Topical clotrimazole for tinea capitis."** Wrong: needs **oral griseofulvin or terbinafine**.
- **"Steroid–antifungal combination cream for a ringworm."** Wrong: causes **tinea incognito**.
- **"Undiluted benzyl benzoate for a toddler."** Wrong: **dilute** (1:3 under 2 years).
- **"Itch persists 1 week after scabies treatment, so it failed."** Wrong: itch can last **2–4 weeks**; look for new burrows.
- **"Soap and frequent baths to keep eczema clean."** Wrong: soap dries the skin; use **emollient as soap substitute**.
- **"Mouth ulcers and rash on cotrimoxazole: add an antihistamine and continue."** Wrong: **SJS**: stop the drug and never rechallenge.
- **"Molluscum in a 9-year-old with 60 lesions: reassure only."** Wrong: **test for HIV**.

Questions seniors ask

How would you describe this rash? Use the SCALE order: site and distribution, colour, arrangement, primary lesion and secondary changes, then extras (mucosa, nails, hair, nodes, systemic signs).

What are the features of scabies in an infant? Irritability, poor sleep, papules, vesicles and pustules on the palms and soles, face and scalp, axillae and nappy area; nodules; burrows may be hard to see.

How do you manage tinea capitis in a school child? Oral griseofulvin with fatty food for 6–8 weeks (or terbinafine 4 weeks), antifungal shampoo for the child and siblings, no sharing of combs, hats or clippers; the child can stay in school once treated.

Why do we care about impetigo beyond the skin? Streptococcal impetigo can cause PSGN 3–6 weeks later (and possibly rheumatic fever); staphylococcal toxins can cause SSSS in infants.

What is your stepped treatment for atopic eczema? Emollients and soap substitutes for everyone; mild steroid for mild flares, moderate for moderate, potent (short courses, not face) for severe; treat infection; sedating antihistamine for sleep; education and follow-up.

How do you care for the eyes in SJS? Ophthalmology review within 24 hours, saline cleaning, lubricants, antibiotic ointment, and separating the lids to prevent adhesions; blindness is a major long-term complication.

What skin changes do you see in kwashiorkor and how do you treat them? Flaky-paint dermatosis with raw, depigmented areas; barrier cream (zinc and castor oil or petroleum jelly), potassium permanganate 0.01% soaks, gentian violet or nystatin for sores, as part of the full SAM protocol with zinc in the feeds.

SELF-TEST

Flashcards

30 cards. Cover the right-hand column and test yourself.

#	Question	Answer
1	Macule vs papule vs plaque?	**Macule**: flat colour change <1 cm. **Papule**: raised solid lesion <0.5–1 cm. **Plaque**: raised flat-topped lesion >1 cm.
2	Vesicle vs bulla vs pustule vs wheal?	**Vesicle**: fluid-filled <0.5 cm. **Bulla**: fluid-filled >0.5 cm. **Pustule**: pus-filled. **Wheal**: transient itchy oedematous plaque lasting <24 h (urticaria).
3	The SCALE order for describing a rash?	**S**ite/distribution, **C**olour, **A**rrangement, **L**esion type (primary then secondary), **E**xtras (mucosa, nails, hair, nodes, systemic features).
4	Typical sites of scabies burrows in a child, and in an infant?	Child: **finger webs, wrists, elbows, anterior axillary folds, nipples, umbilicus/waist, buttocks, genitals**. Infant: also **palms, soles, face and scalp**.
5	Why does scabies itch start weeks after infestation and continue after treatment?	The itch is a **delayed hypersensitivity** to mites, eggs and faeces: starts **3–6 weeks** after first infestation and can last **2–4 weeks** after successful treatment.
6	Permethrin 5% for scabies: how to use?	From **2 months**: apply **neck to toe** (whole body including scalp and face in infants, avoiding eyes/mouth), wash off after **8–12 h**, **repeat after 7 days**; treat all contacts at the same time.
7	Benzyl benzoate 25% dilutions for children (UCG 2023)?	**2–12 years: 1 part lotion + 1 part water. Under 2 years: 1 part lotion + 3 parts water.** Apply at bedtime to the whole body, wash off in the morning, repeat.
8	Ivermectin for scabies?	**200 µg/kg orally** with food, **two doses 7 days apart**; children **≥15 kg**; not in pregnancy. Used for crusted scabies (with topical), outbreaks and topical failure.
9	General measures in scabies?	Treat **all household members and close contacts simultaneously**; wash bedding/clothes/towels in hot water and sun-dry or seal in a bag for 72 h; cut nails; antihistamine at night.
10	What is crusted (Norwegian) scabies?	Hyperinfestation with thousands of mites causing thick crusts, often little itch, in **HIV, malnutrition, steroids or disability**; highly contagious. Ivermectin (multiple doses) + topical scabicide + keratolytic; isolate; HIV test.
11	Why must tinea capitis be treated orally?	The dermatophyte lives **inside the hair shaft and follicle**, which topical antifungals cannot reach (UCG 2023).
12	Griseofulvin for tinea capitis?	**UCG 2023: 10 mg/kg/day** once daily after meals for **6 weeks** (double in severe infection); international guides **15–25 mg/kg/day** for 6–8 weeks; give **with fatty food**.
13	Terbinafine weight bands for tinea capitis?	**<20 kg: 62.5 mg; 20–40 kg: 125 mg; >40 kg: 250 mg** once daily for **4 weeks** (best for *Trichophyton*).
14	What is a kerion and how is it managed?	A boggy, pustular, tender inflammatory mass of **tinea capitis**; treat with **oral antifungal** ± short oral steroid; **do not incise** (not an abscess).
15	Treatment of tinea corporis (UCG 2023)?	**Whitfield's ointment** (benzoic + salicylic acid) 12-hourly until 2 weeks after clearing, or **clotrimazole 1% / miconazole 2%** twice daily for 2–3 weeks; oral griseofulvin if extensive or failing.

#	Question	Answer
16	What is tinea incognito?	Tinea treated with a **steroid cream** (often an over-the-counter combination cream): inflammation is suppressed, the fungus spreads and the rash loses its typical appearance.
17	Non-bullous vs bullous impetigo?	**Non-bullous**: honey-coloured crusts around nose/mouth (*S. aureus* ± *S. pyogenes*). **Bullous**: flaccid blisters leaving collarettes of scale, caused by **staphylococcal exfoliative toxin**.
18	Impetigo treatment (UCG 2023)?	**<5 lesions**: clean with chlorhexidine 0.05%, **gentian violet 0.5% 12-hourly for 3 days** or mupirocin 2% 8-hourly for 5–7 days. **>5 lesions, fever or spread**: **cloxacillin 12.5–25 mg/kg 6-hourly for 7 days** (erythromycin 7.5 mg/kg 6-hourly if allergic).
19	Which renal complication follows streptococcal skin infection, and when?	**Post-streptococcal glomerulonephritis** about **3–6 weeks** after impetigo/infected scabies (1–2 weeks after pharyngitis): haematuria, oedema, hypertension, oliguria.
20	Stepped treatment for atopic eczema?	**Emollients and soap substitute for all**; mild: hydrocortisone 1%; moderate: moderate-potency steroid (e.g. clobetasone 0.05%); severe: potent (betamethasone 0.1%) short courses, not face/genitals; sedating antihistamine at night; treat infection.
21	What is a fingertip unit?	Cream squeezed from a 5 mm nozzle along an adult index finger from tip to first crease: about **0.5 g**, enough for an area of **two adult palms**.
22	Eczema herpeticum: features and treatment?	Painful, clustered **punched-out erosions** on eczematous skin with fever (HSV). **Aciclovir** 20 mg/kg (max 400 mg) orally 5 times daily, or **IV 10–20 mg/kg 8-hourly** if unwell or near the eye; eye review.
23	Molluscum contagiosum: appearance, course and HIV clue?	Pearly **umbilicated papules** 1–5 mm; poxvirus; self-limiting over **6–18 months**. **Extensive, giant or facial molluscum in an older child → test for HIV**.
24	Cetirizine and chlorphenamine doses for children (UCG 2023)?	**Cetirizine**: 1–6 years 2.5 mg, ≥6 years 5 mg once daily. **Chlorphenamine**: 1–2 years 1 mg 12-hourly; 2–5 years 1 mg 6-hourly; 5–12 years 2 mg 6-hourly.
25	What is papular urticaria?	Insect-bite hypersensitivity (mosquitoes, fleas, bedbugs): crops of itchy papules with a central punctum on exposed skin in young children; bed nets, treat bedbugs/fleas, antihistamine, short mild steroid.
26	SJS vs overlap vs TEN by body surface area?	**SJS <10%**, **SJS/TEN overlap 10–30%**, **TEN >30%** of body surface with epidermal detachment; mucosal involvement in all.
27	Commonest SJS triggers in Ugandan children?	**Cotrimoxazole** (and other sulfonamides), **nevirapine**, **anticonvulsants** (carbamazepine, phenytoin, phenobarbital, lamotrigine), **anti-TB drugs**, NSAIDs, allopurinol; Mycoplasma infection.
28	Key management steps in SJS/TEN?	**Stop the drug**, admit, fluids like a burn, non-adherent dressings, **eye review within 24 h**, mouth/genital care, nutrition, analgesia, treat proven infection; **no routine steroids**; never rechallenge; switch cotrimoxazole prophylaxis to **dapsone 2 mg/kg daily**.
29	WHO clinical stage 2 skin/mucosal conditions in children?	**Papular pruritic eruption**, extensive molluscum, extensive wart virus infection, fungal nail infection, **herpes zoster**, recurrent oral ulcers, lineal gingival erythema, angular cheilitis.
30	Skin care in kwashiorkor (WHO)?	Zinc (in feeds); **0.01% potassium permanganate** soaks 10 min daily; **barrier cream** (zinc and castor oil, petroleum jelly, tulle gras) to raw areas; gentian violet or nystatin to sores; avoid nappies.

SELF-TEST

Single best answer questions

15 questions. The answers and explanations follow at the end.

1. A 7-year-old boy (22 kg) has 4 weeks of itching that keeps him awake. His mother and two siblings also itch. There are excoriated papules and burrows in the finger webs, wrists and around the umbilicus and penis. What is the MOST appropriate management?

A Betamethasone 0.1% cream to the itchy areas for 2 weeks

B Oral griseofulvin 10 mg/kg/day for 6 weeks

C Treat only the boy with permethrin and review in 1 month

D Oral cloxacillin and chlorphenamine

E Treat all household members at the same time with a scabicide (e.g. permethrin 5% or diluted benzyl benzoate), repeat after 7 days, wash bedding, and give an antihistamine at night

2. A mother brings her 18-month-old daughter (10 kg) who has scabies. Only benzyl benzoate 25% lotion is available at the HC III. How should it be used for this child (UCG 2023)?

A Undiluted, applied to the whole body, left for 3 days

B Diluted 1 part lotion with 3 parts water, applied to the whole body (avoiding eyes and mouth) overnight, then washed off

C Diluted 1:1 with water and applied only to the hands

D Taken orally, 5 mL twice daily

E Not used; give oral ivermectin 2 mg instead

3. Two weeks after correct permethrin treatment of her whole family, a 6-year-old girl still itches at night. There are healing excoriations but no new burrows. What is the BEST response? Advanced

A Treatment has failed; give oral ivermectin to all

B Start a potent steroid for 4 weeks

C Diagnose atopic eczema and start emollients only

D Reassure that itch can persist 2–4 weeks after successful treatment; give an antihistamine and emollient and review in 2 weeks

E Admit for crusted scabies

4. A 7-year-old girl (20 kg) has had a scaly patch with broken hairs and hair loss on her scalp for 2 months; two classmates have similar patches. A clotrimazole cream did not help. What is the MOST appropriate treatment?

A Oral griseofulvin with fatty food for 6–8 weeks (UCG 10 mg/kg/day, doubled if severe), plus antifungal shampoo

B Continue clotrimazole cream for another month

C Betamethasone 0.1% scalp application

D Oral cloxacillin for 7 days

E Shave the head and apply gentian violet only

5. A 10-year-old has a 4 cm ring-shaped lesion on the thigh that has grown for 2 months despite a pharmacy cream containing betamethasone, clotrimazole and gentamicin. It now looks red and poorly defined with pustules at the edge. What is the MOST likely explanation? Advanced

A Psoriasis
B Contact dermatitis to gentamicin
C Tinea incognito: steroid has modified and spread a tinea infection
D Impetigo
E Granuloma annulare

6. A 4-year-old (16 kg) has 12 honey-crusted lesions around the nose, mouth and on both legs, with tender inguinal nodes and temperature 38.2 °C. Which treatment is MOST appropriate?

A Gentian violet 0.5% alone for 3 days
B Topical betamethasone
C Oral cloxacillin 250 mg (about 15 mg/kg) 6-hourly for 7 days, with crust removal and hygiene
D Oral griseofulvin
E Oral amoxicillin 125 mg once daily

7. A 6-year-old boy treated for impetigo 3 weeks ago now has puffy eyelids, cola-coloured urine and BP 130/88. What is the MOST likely diagnosis?

A Nephrotic syndrome
B Staphylococcal scalded skin syndrome
C Allergic reaction to cloxacillin
D Urinary tract infection
E Post-streptococcal glomerulonephritis

8. An 8-month-old has dry, red, itchy patches on the cheeks and extensor surfaces of the limbs, waking at night. His mother washes him with bar soap three times a day. No infection. What is the MOST important first-line advice?

A Stop breastfeeding and start soya formula
B Use emollients (e.g. petroleum jelly) liberally several times a day, stop soap and use an emollient as soap substitute, with hydrocortisone 1% to red areas for flares
C Betamethasone 0.1% to the face twice daily for 4 weeks
D Oral cloxacillin for 7 days
E Oral prednisolone 1 mg/kg for 2 weeks

9. A 3-year-old with long-standing eczema develops fever and painful, clustered, punched-out erosions on the face and neck over 2 days. What is the BEST management? Advanced

A Aciclovir (IV 10–20 mg/kg 8-hourly if unwell or near the eye, otherwise oral 20 mg/kg 5 times daily) and urgent eye review if periocular
B Increase to a potent topical steroid
C Oral cloxacillin alone
D Calamine lotion and reassurance
E Wet wraps with betamethasone

10. A 9-year-old girl on cotrimoxazole prophylaxis (started 2 weeks ago after an HIV diagnosis) presents with fever, painful red eyes, lip crusting, mouth ulcers and dusky macules with blisters covering about 8% of her body. What is the MOST appropriate action?

A Continue cotrimoxazole and add chlorphenamine

B Stop cotrimoxazole and give prednisolone 1 mg/kg at home

C Admit, but restart cotrimoxazole after 1 week

D Stop cotrimoxazole, admit for supportive burn-type care with fluids, eye and mouth care, ophthalmology review, and later switch prophylaxis to dapsone

E Treat as measles with vitamin A

11. A 10-year-old boy has more than 50 pearly, umbilicated papules on his face, neck and trunk, some over 1 cm. He has had recurrent pneumonia. What is the MOST important next step?

A Reassure; molluscum always resolves

B Curette all lesions today

C Apply a potent steroid cream

D HIV test, and if positive start ART (lesions often regress on ART)

E Oral aciclovir

12. A 2-year-old has crops of intensely itchy papules with central puncta on his legs and forearms, recurring for months, especially after visiting his grandmother's house where there are bedbugs. Siblings are less affected; no burrows. What is the MOST likely diagnosis?

A Papular urticaria (insect-bite hypersensitivity)

B Scabies

C Atopic eczema

D Chickenpox

E Papular pruritic eruption of HIV

13. A 5-year-old (18 kg) develops wheals all over his body 30 minutes after eating groundnuts, with lip swelling, hoarseness and wheeze. SpO_2 92%. What is the FIRST treatment?

A Cetirizine 5 mg orally

B Prednisolone 18 mg orally

C Chlorphenamine 2 mg orally

D Calamine lotion

E Adrenaline 1:1000, 0.18 mL IM into the anterolateral thigh

14. A 22-month-old (7.5 kg) with bilateral pitting oedema has dark, cracked, peeling patches on the legs and buttocks with raw pink areas underneath. What is the MOST appropriate skin care alongside the SAM protocol?

A Betamethasone 0.1% ointment to all lesions

B Daily bathing with bar soap and nappies to keep the area clean

C Potassium permanganate 0.01% soaks, barrier cream (zinc and castor oil or petroleum jelly) to raw areas, gentian violet or nystatin for sores, no nappies

D Oral griseofulvin for 6 weeks

E Benzyl benzoate to the whole body

15. A 3-week-old baby has fever, irritability and tender, red skin that peels off in sheets around the mouth, neck folds and groin when gently rubbed. The mouth and eyes are normal. What is the MOST likely diagnosis and treatment? Advanced

- A Toxic epidermal necrolysis; stop all drugs and give steroids
- B Staphylococcal scalded skin syndrome; admit, IV cloxacillin/flucloxacillin, fluids and gentle skin care
- C Bullous impetigo; topical gentian violet only
- D Neonatal eczema; emollients
- E Congenital syphilis; benzathine penicillin IM once

Answers & explanations

1. E · Treat all household members at the same time with a scabicide (e.g. permethrin 5% or diluted benzyl benzoate), repeat after 7 days, wash bedding, and give an antihistamine at night

E is correct: nocturnal itch, burrows in typical sites and affected family members = **scabies**; everyone must be treated **simultaneously** with a repeat application, plus bedding hygiene. A is wrong: steroids relieve itch but do not kill mites and may cause crusted scabies. B is wrong: griseofulvin treats fungal infection. C is wrong: untreated contacts re-infest the child. D is wrong: there is no secondary infection described, and antibiotics do not treat scabies.

2. B · Diluted 1 part lotion with 3 parts water, applied to the whole body (avoiding eyes and mouth) overnight, then washed off

B is correct: under 2 years UCG dilutes benzyl benzoate **1:3 with water**, applied to the whole body (including scalp in infants, avoiding eyes) and washed off after about 12 hours. A is wrong: undiluted lotion is very irritant in young children. C is wrong: 1:1 is for ages 2–12, and the whole body must be treated. D is wrong: benzyl benzoate is topical only. E is wrong: ivermectin is not recommended under 15 kg.

3. D · Reassure that itch can persist 2–4 weeks after successful treatment; give an antihistamine and emollient and review in 2 weeks

D is correct: post-scabies itch is a **hypersensitivity reaction** that can last 2–4 weeks; with no new burrows, reassure, use antihistamine/emollient and review. A is wrong: failure is suggested by new burrows or itching beyond 4 weeks. B is wrong: a prolonged potent steroid is unnecessary. C is wrong: there is no evidence of eczema. E is wrong: crusted scabies has thick crusts and occurs in immunosuppressed children.

4. A · Oral griseofulvin with fatty food for 6–8 weeks (UCG 10 mg/kg/day, doubled if severe), plus antifungal shampoo

A is correct: **tinea capitis** needs **oral** antifungal (griseofulvin, or terbinafine 125 mg/day for 20–40 kg for 4 weeks), with a shampoo to reduce spore shedding. B is wrong: topicals cannot reach the hair follicle. C is wrong: steroids worsen fungal infection. D is wrong: this is not bacterial. E is wrong: gentian violet does not treat the fungus in the hair shaft.

5. C · Tinea incognito: steroid has modified and spread a tinea infection

C is correct: steroid in combination creams suppresses inflammation and allows dermatophytes to spread, producing an atypical, extensive **tinea incognito**; stop the steroid, do a KOH scraping and give an antifungal (oral if extensive). A is wrong: psoriasis gives well-demarcated silvery plaques on extensor surfaces. B is wrong: contact dermatitis would not be annular with an active pustular edge. D is wrong: impetigo gives honey crusts, not an expanding ring. E is wrong: granuloma annulare is non-scaly and non-pustular.

6. C · Oral cloxacillin 250 mg (about 15 mg/kg) 6-hourly for 7 days, with crust removal and hygiene

C is correct: widespread impetigo (>5 lesions) with fever and lymphadenitis needs **systemic anti-staphylococcal treatment**: cloxacillin 12.5–25 mg/kg 6-hourly for 7 days (UCG). A is wrong: topical antiseptics suit localised disease only. B is wrong: steroids worsen infection. D is wrong: this is bacterial, not fungal. E is wrong: amoxicillin does not cover penicillinase-producing *S. aureus*, and once daily is the wrong frequency.

7. E · Post-streptococcal glomerulonephritis

E is correct: haematuria, oedema and hypertension **3–6 weeks after streptococcal skin infection** is classic **PSGN**. A is wrong: nephrotic syndrome causes heavy proteinuria and generalised oedema without cola urine or hypertension typically. B is wrong: SSSS is a skin condition of infants. C is wrong: drug allergy gives rash or urticaria, not nephritis on this timeline. D is wrong: UTI does not cause periorbital oedema and hypertension.

8. B · Use emollients (e.g. petroleum jelly) liberally several times a day, stop soap and use an emollient as soap substitute, with hydrocortisone 1% to red areas for flares

B is correct: **atopic eczema** management rests on **emollients and soap avoidance**, with a mild steroid (hydrocortisone 1%) for facial flares. A is wrong: stopping breastfeeding is not recommended; food allergy is an uncommon cause. C is wrong: potent steroids on an infant's face cause atrophy. D is wrong: there are no signs of infection. E is wrong: systemic steroids are not used for routine eczema.

9. A · Aciclovir (IV 10–20 mg/kg 8-hourly if unwell or near the eye, otherwise oral 20 mg/kg 5 times daily) and urgent eye review if periocular

A is correct: punched-out painful erosions on eczema with fever = **eczema herpeticum** (HSV), an emergency treated with **aciclovir**. B is wrong: steroids worsen herpes infection. C is wrong: antibiotics may be added for secondary infection but do not treat HSV. D is wrong: this is dangerous to dismiss. E is wrong: occlusive steroid wraps would spread the virus.

10. D · Stop cotrimoxazole, admit for supportive burn-type care with fluids, eye and mouth care, ophthalmology review, and later switch prophylaxis to dapsone

D is correct: mucosal involvement of eyes and mouth plus blistering <10% BSA after a new sulfonamide = **SJS**; stop the drug, admit for supportive care with eye review, never rechallenge, and use **dapsone** for PCP prophylaxis. A is wrong: continuing the culprit drug can progress to TEN. B is wrong: SJS needs admission and steroids are not routinely recommended (UCG). C is wrong: rechallenge can be fatal. E is wrong: measles has a morbilliform rash without blistering and the drug history points to SJS.

11. D · HIV test, and if positive start ART (lesions often regress on ART)

D is correct: **extensive and giant molluscum** in an older child with recurrent pneumonia strongly suggests **HIV** (WHO stage 2 sign); ART is the most effective treatment. A is wrong: reassurance misses the underlying immunosuppression. B is wrong: destroying lesions without addressing the cause leads to recurrence. C is wrong: steroids worsen viral skin infections. E is wrong: aciclovir does not act on poxvirus.

12. A · Papular urticaria (insect-bite hypersensitivity)

A is correct: grouped itchy papules with puncta on exposed skin, recurring after insect exposure, without burrows = **papular urticaria**. B is wrong: scabies has burrows in webs/wrists/genitals and affects contacts. C is wrong: eczema is dry and flexural (or facial in infants), not punctate. D is wrong: chickenpox is an acute febrile vesicular illness. E is wrong: PPE is possible but HIV should be considered only if there are other clues; the history here fits insect bites.

13. E · Adrenaline 1:1000, 0.18 mL IM into the anterolateral thigh

E is correct: urticaria with **airway (hoarseness, lip swelling) and breathing (wheeze, hypoxaemia)** involvement is **anaphylaxis**: IM adrenaline 0.01 mL/kg of 1:1000 (UCG band <6 years 0.15 mL), repeated every 5 minutes, plus oxygen. A, B and C are wrong: antihistamines and steroids are second-line and too slow to treat airway swelling. D is wrong: calamine is only for itch.

14. C · Potassium permanganate 0.01% soaks, barrier cream (zinc and castor oil or petroleum jelly) to raw areas, gentian violet or nystatin for sores, no nappies

C is correct: this is **kwashiorkor dermatosis**; WHO recommends potassium permanganate soaks, barrier creams, gentian violet/nystatin for sores and leaving the perineum uncovered, with zinc in the feeds. A is wrong: steroids thin damaged skin and increase infection. B is wrong: soap dries the skin and nappies keep raw areas moist. D is wrong: this is not a fungal infection. E is wrong: there is no scabies.

15. B · Staphylococcal scalded skin syndrome; admit, IV cloxacillin/flucloxacillin, fluids and gentle skin care

B is correct: superficial peeling with a positive Nikolsky sign, flexural and perioral accentuation, fever and **sparing of mucous membranes** in an infant = **SSSS** (staphylococcal exfoliative toxin); treat with IV anti-staphylococcal antibiotics, fluids and burn-type skin care. A is wrong: TEN involves mucosae and usually follows a drug; steroids are not indicated. C is wrong: bullous impetigo is localised and the baby is well. D is wrong: eczema does not peel in sheets with fever. E is wrong: congenital syphilis can blister palms and soles, but needs a full course of IV benzylpenicillin, not a single benzathine dose.

SELF-TEST

Clinical cases

Work through each step before reading the model answer.

Case 1: A household with itch and sores

A mother brings her 6-year-old son (18 kg) to the Mengo outpatient clinic: he has itched for 3 weeks, mostly at night, and has had painful crusted sores on his hands for 5 days. His sisters aged 4 and 18 months and the mother also itch. They share one bed in a single room in Kisenyi.

1. What will you look for on examination?

> Distribution and type of lesions: **burrows and excoriated papules** in finger webs, wrists, axillae, umbilicus, genitals (and palms/soles/scalp in the baby); **secondary infection** (honey crusts, pustules, cellulitis, nodes, fever); general state, nutrition; **BP and urine dipstick** for PSGN; signs of HIV if relevant. He has burrows in typical sites and honey-crusted lesions on both hands; BP and urine normal.

2. What is your diagnosis?

> **Scabies** with **secondary impetigo**, household outbreak.

3. Prescribe for the family.

> Scabicide for **all four at the same time, the same night**: permethrin 5% (from 2 months) or **benzyl benzoate 25%: full strength for the mother, diluted 1:1 for the 6- and 4-year-olds, 1:3 for the 18-month-old**; neck to toe (whole body including scalp in the baby), wash off in the morning, **repeat** (after 7 days with permethrin). For the impetigo (>5 lesions): **cloxacillin 250 mg 6-hourly for 7 days** (≈14 mg/kg). Chlorphenamine 2 mg at night for the boy.

4. What practical advice do you give the mother?

> Wash all bedding, clothes and towels in hot water and dry in the sun (or seal in a bag for 72 h) on the day of treatment; cut nails short; reapply to hands if washed; **itch can last 2–4 weeks**; come back if new sores, facial swelling or dark urine (PSGN).

5. At review in 3 weeks the boy has no new burrows but the 18-month-old has new papules on her soles. What now?

> Likely **treatment failure or re-infestation in the baby** (possibly under-applied to soles/scalp, or another contact untreated, e.g. a visiting relative). Re-treat the whole household correctly, check all contacts; ivermectin is not used under 15 kg, so repeat permethrin/diluted benzyl benzoate or sulphur ointment for 3 nights.

Take-away: Scabies is a household disease: treat everyone at once, repeat, clean bedding, treat secondary impetigo, and remember PSGN.

Case 2: Rash and mouth sores on cotrimoxazole

A 9-year-old girl (24 kg), newly diagnosed with HIV 3 weeks ago, was started on cotrimoxazole prophylaxis and is due to start ART next week. She now has 3 days of fever, sore red eyes, painful mouth ulcers so she cannot eat, and dusky red macules on the trunk with a few flaccid blisters.

1. What is the likely diagnosis and how do you estimate severity?

> **Stevens–Johnson syndrome** triggered by **cotrimoxazole** (onset 4–28 days after a new drug, fever, ≥2 mucosae, blistering). Estimate % BSA with **epidermal detachment** (child's palm ≈1%): about 6% → **SJS** (<10%); overlap 10–30%; TEN >30%. Check the Nikolsky sign and vitals.

2. What are your immediate actions?

> **Stop cotrimoxazole** (and any non-essential drug), document the allergy clearly, **admit**; IV access in unaffected skin, fluids (maintenance plus losses; urine output ≥1 mL/kg/h), analgesia (paracetamol, morphine), warm room, barrier nursing.

3. How do you care for her skin, eyes and mouth?

> **Skin**: leave blisters intact, non-adherent dressings (paraffin gauze, petroleum jelly), no adhesive tape. **Eyes**: **ophthalmology within 24 h**, saline cleaning, lubricants, antibiotic ointment, separate the lids. **Mouth**: saline/chlorhexidine mouthwash, petroleum jelly to lips, analgesia before feeds; NG feeding if needed. Genital care to prevent adhesions.

4. Would you give steroids or prophylactic antibiotics?

> **No routine steroids** (UCG 2023: no strong evidence; they increase infection and catabolism) and **no prophylactic antibiotics**: monitor and culture, treat proven infection.

5. What changes in her HIV care after recovery?

> **Never rechallenge** with cotrimoxazole or other sulfonamides; PCP prophylaxis with **dapsone 2 mg/kg once daily (max 100 mg)**; start ART (DTG-based per Uganda guidelines, avoiding nevirapine); record the reaction in her HIV card; report the adverse drug reaction.

> **Take-away:** Fever, sore eyes and mouth, and a blistering rash after a new drug = SJS: stop the drug, burn-type supportive care, early eye review, never rechallenge.

Case 3: The tutorial case: an itchy infant with weeping cheeks

For the skin tutorial, Edith presents an 8-month-old boy (7.8 kg, WAZ −0.6) with 3 months of itchy, red, dry patches on both cheeks, scalp and the outer arms and shins, worse in the last 3 days with yellow crusts and weeping on the cheeks. He is exclusively breastfed plus porridge, immunised up to date for age (measles-rubella is due at 9 months), and both parents are HIV-negative. His father has asthma.

1. Describe the rash using correct terms and give the diagnosis.

> Symmetrical, **ill-defined erythematous, dry, scaly patches and plaques** on the **cheeks, scalp and extensor limbs** with **excoriations**; on the cheeks **weeping and honey-coloured crusts**. Diagnosis: **infantile atopic eczema** (family history of atopy) **with secondary bacterial infection (impetiginised eczema)**.

2. What must you exclude because it is an emergency?

> **Eczema herpeticum**: painful, clustered, **punched-out erosions**, fever, unwell baby. Here the lesions are crusted and weeping without punched-out erosions, and he is afebrile, so it is bacterial; if any doubt, start aciclovir.

3. Prescribe his treatment.

> **Flucloxacillin/cloxacillin 12.5–25 mg/kg 6-hourly** (about 125 mg 6-hourly) for 7 days; **emollient** (petroleum jelly or emulsifying ointment) liberally 3–4 times a day and as a **soap substitute**; **hydrocortisone 1%** once or twice daily to red areas on the face and body for 7–14 days (FTU guidance: about 1 FTU for the face and neck at this age); stop bar soap; cotton clothes; antihistamine generally avoided at this age (promethazine contraindicated under 2 years).

4. How would you explain the condition to the mother?

> Eczema is a **dry, sensitive skin** condition that runs in families with asthma/allergy; it comes and goes and usually improves as children grow; it is **not contagious**; the most important treatment is **daily moisturiser**, even when the skin looks good; steroid creams are safe when used correctly in small amounts for short periods; come back urgently if painful sores or fever appear.

5. Summarise this case for the group in two sentences as you would in the tutorial.

> "An 8-month-old with infantile atopic eczema of the face and extensors, now secondarily infected with Staph, in a family with asthma. Treatment is oral cloxacillin, emollients as both moisturiser and soap substitute, and hydrocortisone 1% for flares, with a clear warning about eczema herpeticum."

Take-away: Describe, then diagnose; emollients are the backbone of eczema care, infection needs anti-staphylococcal antibiotics, and punched-out erosions mean herpes.

References

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A study aid, not a protocol. Check doses against the Uganda Clinical Guidelines 2023, your unit's protocol and a senior before you treat a patient.